

SPHERE: MISSION CONTROL

TEACHER PEDAGOGICAL DEPLOYMENT MANUAL

DOCUMENT REF // EU_SPHERE_INSTRUCTOR_V2

Instructor Overview: This document maps the SPHERE Thermal Suit experiment to standard European secondary physics frameworks. It includes lesson timings, safety protocols, expected analytical constants, and suggestions for discussion to streamline execution in single-session classroom windows.

1. CURRICULAR ALIGNMENT & LESSON SCHEME

This curriculum is aligned with the **Inquiry-Based Science Education (IBSE)** pedagogy, specifically mimicking the virtual-physical calibration workflows of the **European Union Horizon 2020 SPHERE initiative**. It directly maps to secondary physics curriculum segments covering:

- **Thermodynamics:** Thermal conduction, convection, radiation, thermal barriers, and specific heat capacity calculations.
- **Mathematical Modeling:** Rate equations, Newtonian decay curves, and parametric modeling.
- **Digital Twin Concepts:** Students use a simplified computer model to compare predicted and measured results.

SUGGESTED LESSON STRUCTURE (45-60 MINUTES):

TIMING	INQUIRY STEP	STUDENT ACTIVITIES & MILESTONES
00 - 10 Min	Engage & Certify	Students review the deep-space context, read the student briefing, and complete the Pre-Flight Quiz to unlock mission telemetry.
10 - 20 Min	Explore & Construct	Student crews construct one of four thermal protection systems: • Bare Capsule • Cotton Layer • Mylar Layer • Mylar + Cotton + Bubble Wrap They then install the temperature probe through the PG7 cable gland and prepare the capsule for testing.
20 - 25 Min	Model & Predict	Crews open the Simulator tab, select their insulation schemes, configure parameters ($T_0 = 80^{\circ}\text{C}$), and execute mathematical predictions.
25 - 40 Min	Explain & Measure	Pour 80°C water into PET jars. Submerge in the 0°C ice baths. Start stopwatch. Students log temperatures every 60 seconds.
40 - 45 Min	Elaborate & Grade	Analyze the resulting Thermal Correlation Score . Discuss convective leaks, and record grading coefficients.

2. RISK ASSESSMENT & HAZARD MITIGATION

Enforce these standard safety mandates to guarantee zero accident rates during live laboratory sessions:

HAZARD	LEVEL	MANDATED MITIGATION STEPS
Hot Water Scalding	HIGH	Do not allow students to use water > 80°C. Teachers should pre-boil and dispense water pre-mixed with cold tap water to lock thermal limits at exactly 80°C.
Melted Ice Slippages	MEDIUM	Place physical ice baths in high-density plastic trays. Provide moisture absorbent towels. Keep ice bath tables completely segregated from laptops and tablets.
Electrical Shock	MEDIUM	Only permit low-voltage, USB-powered or battery-powered digital sensor readouts. Keep the thermocouple cables long to maximize distance from the computer screen.
Capsule Deformation	LOW	Ensure thin PET food jars are not exposed to boiling 100°C temperatures which causes polymer warping. Enforce the strict 80°C limit.

Summer & High-Ambient Temperature Adjustments (~30°C):

High ambient temperatures in summer classrooms can accelerate ice-bath degradation and introduce safety risks. Enforce these pedagogical workarounds:

- 1. Thermal Baseline Safety Scaling:** If ambient room temperatures are high, reduce the hot water target from **80°C to 55°C–60°C**. This eliminates scalding risks and thermal stress on PET caps while preserving perfect exponential Newtonian cooling curves. Students simply adjust their initial temperature parameter ($T_o = 60^{\circ}\text{C}$) inside the digital twin simulator to maintain flawless correlation.
- 2. Ice Bath Melt Compensation:** In hot rooms, the ice bath temperature may rise to 3°C–6°C due to melting. Instruct students to measure the actual bath temperature with their probe at the start of the re-entry drop, and configure the environmental temperature (T_{env}) in both the simulator and chart overlays to match this observed value (e.g. $T_{env} = 4.5^{\circ}\text{C}$).
- 3. Ice-Free Alternative Setup:** If ice is logistically unavailable, students can use ambient room tap water (typically ~25°C–30°C) as their cooling environment. To simulate deep-space decay, configure the digital twin simulator with $T_{env} = 30^{\circ}\text{C}$ (or the measured tap water temperature) and $T_o = 60^{\circ}\text{C}$. The mathematical model behaves identically, allowing valid heat transfer comparisons.

3. ANALYTICAL SOLUTIONS KEY (NEWTONIAN DECAY CONSTANTS)

Crews configure their spacesuits with distinct insulation wraps. The following values are simplified educational estimates intended to demonstrate the relative effectiveness of different insulation strategies.

Important: Actual classroom measurements may differ depending on wrapping quality, material thickness, sensor placement, room conditions, and ice-bath temperature.

Important Experimental Design Note:
Students do not test bubble wrap as a standalone insulation system. Bubble wrap is only used as an outer protective layer in the combined configuration (Mylar + Cotton + Bubble Wrap). This reflects real engineering practice, where multiple materials are combined to reduce different heat-transfer mechanisms simultaneously.

EXPECTED COOLING RATES AFTER 15-MINUTE SUBMERSION:

INSULATION DESIGN	DECAY CONSTANT (k)	EXPECTED END TEMP (15 MIN)	PRIMARY INSULATION EFFECT
A. Bare Capsule	$k \approx 0.082$	$\approx 23^{\circ}\text{C}$	Baseline control. No additional insulation.
B. Cotton Layer	$k \approx 0.050$	$\approx 37^{\circ}\text{C}$	Reduces conductive heat transfer through trapped air within fibers.
C. Mylar Layer	$k \approx 0.040$	$\approx 43^{\circ}\text{C}$	Reflects thermal radiation and reduces radiative heat loss.
D. Mylar + Cotton + Bubble Wrap	$k \approx 0.022$	$\approx 56^{\circ}\text{C}$	Combined reduction of radiation, conduction and convective losses.

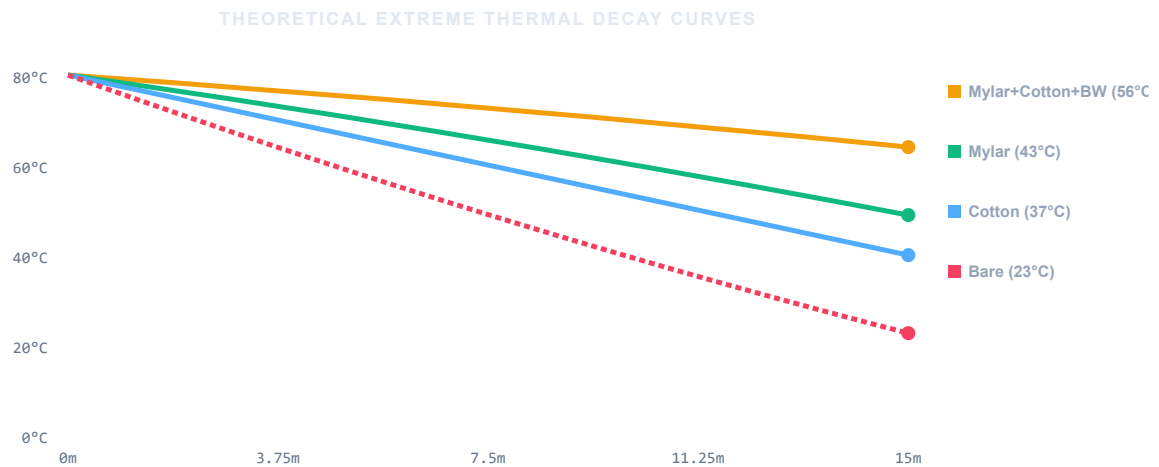


FIGURE 3: EXPECTED EXPONENTIAL COOLING CURVES ($t = 0$ to 15m)

*Note: Physical values may deviate by $\pm 5\%$ depending on wrapping tightness and capsule sealing. The **Thermal Correlation Score** provides a visual comparison between the measured data and the prediction model. It is intended as a discussion tool rather than a measure of success.*

4. CLASSROOM DISCUSSION & REFLECTION

The goal is to develop a scientific understanding and engineering intuition rather than achieving specific results. Teachers are encouraged to let the students discuss and assess their results using the questions below:

- Which type of insulation performed best and why?
- Did your measured temperatures match the simulated prediction? If not, what might explain the differences?
- Which heat transfer mechanisms are dominant in which material?
- How does this experiment differ from the actual conditions in space?
- What limitations does this experimental model have?
- How could the experiment be improved?

Teacher Note: Deviations between prediction and measurement should be viewed as opportunities for scientific discussion rather than as mistakes. Real experiments rarely match theoretical models perfectly.

5. EXPERIMENT LIMITATIONS

This experiment is a simplified model of thermal protection in space and does not perfectly represent real astronaut conditions.

- Real space is a vacuum, while this experiment takes place in air and water.
- Heat transfer in space is dominated by radiation, whereas conduction and convection are significant in this experiment.
- Real spacesuits use many specialized insulation layers and active thermal control systems.
- Astronauts continuously generate heat through metabolism.
- The experiment is designed to illustrate heat transfer concepts rather than accurately reproduce space conditions.